SANGBAY
Gas Sensor

Methane Sensor

N-CH_4 Application & Specification

Infrared CH_4 Sensor

High Sensitivity High Definition
Low Consumption Poison-resistant

Shanghai Sangbay Sensor Technology Co., Ltd

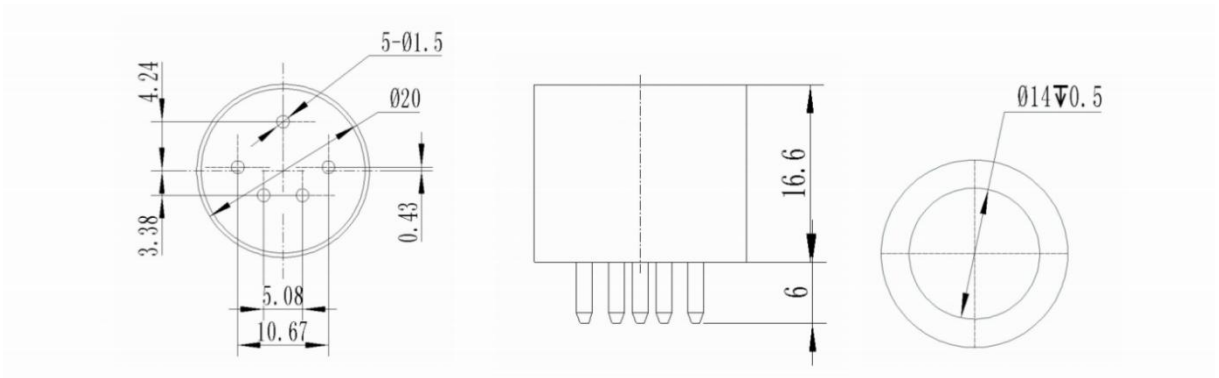
1. Technical Parameters

Model	N-CH_4
Target Gas	CH_4
Working Voltage	3.6~5V DC
Average Current	<60mA
Interface level	3.0V
Signal Output	UART 0.4~2V
Preheating Time	60S
Response Time	T_{90} < 30s
Working Temperature	-20~60 ℃
Working Humidity	0~95% RH (No Condensation)
Overall dimension	Φ20 mmX21.4 mm
Weight	35g
Service Life	>5 Years
Protection Level	IP54
Intrinsically Safe Parameters	Ui: 5.5V, Ii: 125mA Pi: 0.172W, Ci: 10.92μF, Li: 0μH

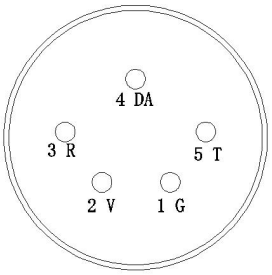
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- **Ranged often used and accuracy**

Gas	Formula	Range	Resolution	Accuracy	Remarks
Methane	CH_4	0-20000 PPM	50 PPM	± (0.05%vol+ 5%READING)	Temperatures Compensation
		0~5.00%Vol	0.01%Vol		
		0~10.00%Vol			
		0~20.00%Vol			
		0~100%%Vol	0.1%Vol		



Name	Description
Pin 2	Vin Voltage input positive
Pin 1	GND Voltage input negative
Pin 4	DA (0.4~2V)
Pin 3	R(RXD) 0~3.0V data input
Pin 5	T(TXD) 0~3.0V data output



- **Output mode** Analog voltage output

DA output voltage range (0.4V~2V), corresponding to gas concentration (0~full scale). Connect the sensor Vin terminal to 5V, GND terminal to power ground, and DA terminal to the input terminal of ADC. The sensor outputs the voltage value representing the gas concentration from the DA terminal after the preheating time.When the self-test fails, the output voltage of the sensor is 0V.

- **Intrinsically safe explosion-proof**

This product complies with GB/T3836.1-2021 Explosive Atmospheres Part 1: General Requirements for Equipment GB/T3836.4-2021 Explosive Atmospheres Part 4: Equipment Protected by Intrinsically Safe "i" and GB3836.20-2010 Explosive Atmospheres Part 20: Equipment with Equipment Protection Level (EPL) Ga; The explosion-proof sign is Ex ia IIC T4 Ga, which is applicable to explosive environment in Zone 0, Zone 1 and Zone 2; It has passed the inspection of the National Quality Inspection Center for Explosion-proof Electrical Products and obtained the explosion-proof certificate. Please pay attention to the following items when using:

1. The intrinsically safe power supply must be used to power the sensor, otherwise the explosion-proof performance will be affected.
2. It is forbidden to replace the sensor in dangerous places.
3. It is forbidden to disassemble and replace sensor elements to avoid affecting explosion-proof performance.
4. It is not allowed to replace components or structures to avoid affecting explosion-proof performance.

Address: Building No.5 of Huiwangdong Road, Jiading District, Shanghai
Tel: +86 18538212056 E-mail: sales@sangbay.com Web: www.sangbay-sensor.com

- **Attention**

1. The sensor should be calibrated regularly, and the recommended calibration period is 6 months.
2. Do not use the sensor for a long time in an environment with high dust density.
3. Please use the sensor within the power supply range of the sensor.
4. It is forbidden to cut or weld sensor pins.
5. Products with different ranges should be selected according to different application fields and occasions.